

J KARTHIK

192472136

CSA0911

JAVA

1. Printer Queue – Inter – Thread Communication

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

J.Karthik
192472136RB

QUESTIONS

Q1
Printer Queue ♦ Inter-Thread Communication
25 marks

Q2
Hospital Resource Allocation ♦ Synchronization and Communication
25 marks

Q3
Bank Withdrawal ♦ Insufficient Balance Exception
25 marks

Q4
Multithreaded Number Processing
25 marks

4/4 answered

Q1 Printer Queue ♦ Inter-Thread Communication 25 marks

Problem Statement:

Develop a Java program to simulate a shared printer used by multiple users. A user thread submits a document while a printer thread processes it.

Requirements:

Create producer and consumer threads.
Use a shared document queue/buffer.
Implement wait() and notify().
Ensure proper communication between the threads.
Display submission and printing status.

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Printer p = new Printer();
5
6         Thread user = new Thread(() -> {
7             for (int i = 1; i <= 5; i++)
8                 try { p.submit("Document " + i); }
9                 catch (Exception e) {}
10        });
11
12        Thread printer = new Thread(() -> {
13            for (int i = 1; i <= 5; i++)
14                try { p.print(); }
15                catch (Exception e) {}
16        });
17
18        user.start();
```

Input

stdin input

Output

Submitted: Document 1
Printing: Document 1
Submitted: Document 2
Printing: Document 2
Submitted: Document 3
Printing: Document 3

2. Hospital Resource Allocation – Synchronization and Communication

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

J.Karthik
192472136RB

CO3-AT3-Assignment

← Back

QUESTIONS**Q1** Printer Queue ♦ Inter-Thread Communication 25 marks**Q2** Hospital Resource Allocation ♦ Synchronization and Communication 25 marks**Q3** Bank Withdrawal ♦ Insufficient Balance Exception 25 marks**Q4** Multithreaded Number Processing 25 marks

4/4 answered

Q2 Hospital Resource Allocation ♦ Synchronization and Communication 25 marks**Problem Statement:**
Develop a Java program to simulate allocation of a limited number of hospital resources such as ventilators beds or medical equipment to multiple patients.**Requirements:**
Create multiple patient/request threads.
Maintain a shared number of available resources.
Use synchronization to control resource allocation.
Use inter-thread communication when resources become unavailable.
Release resources after use and notify waiting threads.
Handle invalid resource requests using appropriate exception handling.**Your Code****Input**
stdin input**Output**
Submitted: Document-1
Submitted: Document-2
Printing: Document-1
Printing: Document-2

3. Bank Withdrawal – Insufficient Exception

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

J.Karthik
192472136RB

CO3-AT3-Assignment

← Back

QUESTIONS**Q1** Printer Queue ♦ Inter-Thread Communication 25 marks**Q2** Hospital Resource Allocation ♦ Synchronization and Communication 25 marks**Q3** Bank Withdrawal ♦ Insufficient Balance Exception 25 marks**Q4** Multithreaded Number Processing 25 marks

4/4 answered

Q3 Bank Withdrawal ♦ Insufficient Balance Exception 25 marks**Problem Statement:**
Develop a Java program to simulate a bank withdrawal operation. If the withdrawal amount exceeds the available balance the program should generate and handle a user-defined InsufficientBalanceException.**Requirements:**
Read account balance and withdrawal amount.
Create a user-defined exception.
Throw the exception when the withdrawal amount exceeds the balance.
Display the updated balance for a successful transaction.**Your Code****Input**
stdin input**Output**
Withdrawal Successful
Balance = 20000.0

4. Multithreaded Number Processing



SIMATS Online Programming Lab
JAVA PROGRAMMING

J.Karthik
192472136RB

QUESTIONS

Q1
Printer Queue 
25 marks

Q2
Hospital Resource Allocation 
25 marks

Q3
Bank Withdrawal 
25 marks

Q4
Multithreaded Number Processing
25 marks

4/4 answered

Q4 Multithreaded Number Processing

25 marks

Problem Statement:

Develop a Java program that creates multiple threads to perform different operations on a set of numbers.

Requirements:

Create separate threads to calculate the sum average and maximum value.
Use the Thread class.
Execute the threads concurrently.
Display the result produced by each thread

Your Code

```
1 class Main {
2     public static void main(String[] args) {
3         int[] a = {10, 20, 30, 40, 50};
4         new Thread() -> {
5             int s = 0;
6             for (int n : a) s += n;
7             System.out.println("Sum = " + s);
8         }).start();
9         new Thread() -> {
10            int s = 0;
11            for (int n : a) s += n;
12            System.out.println("Average = " + (double)s / a.length);
13        }).start();
14        new Thread() -> {
15            int m = a[0];
16            for (int n : a) if (n > m) m = n;
17            System.out.println("Maximum = " + m);
18        }).start();
19    }
20 }
```

Input

stdin input

Output

Maximum = 50
Sum = 150
Average = 30.0